

Low-Penalty Joint Communication and Acoustic Sensing using P₄₃ Polyphase Code

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Abstract Polyphase coding is investigated for distributed acoustic sensing in joint communication-sensing systems. A P₄₃ code is evaluated using CC-OTDR and compared to linear frequency-modulated probes. Experiments demonstrate an additional 17.4 dB crosstalk suppression, while inducing only a 0.09 dB OSNR penalty on co-propagating DP-16QAM signals. ©2026 The Author(s)

Introduction

Distributed acoustic sensing based on phase-sensitive optical time domain reflectometry (Φ -OTDR) has been proven as a complementary technique to conventional monitoring technologies in various applications such as intrusion detection [1] and monitoring of critical infrastructure health [2-3], thereby enhancing infrastructure availability.

In a conventional pulse-based Φ -OTDR, the backpropagating signal from Rayleigh backscattering has a low signal-to-noise ratio (SNR) after certain kilometres due to the weak backscattering coefficient in a standard single-mode fibre. In order to increase the SNR, the pulse width of the probe can be broadened, while at the same time, the spatial resolution is reduced. A more sophisticated technique used in radio detection and ranging (RADAR) is pulse compression coding, which increases the SNR while maintaining spatial resolution [4]. However, pulse compression coding introduces sidelobes that spread across the fibre length corresponding to the duration of the probe sequence. Sidelobe noise superimposes at reflections and with backscattering, leading to inter-spatial crosstalk, which is a key performance metric for acoustic sensing interrogators [5]. In [6], Golay pairs were used, but this approach has a reduced detectable event bandwidth due to the larger measurement time. In [7], cycle PRBS was utilised, but it works only for short fibres since the code sequence needs to be longer than the round-trip time of the fibre. In [8],

this issue is addressed using an optimised Frank code; however, the experimental validation was performed on a short fibre incorporating a weak fibre Bragg grating. In [9], this issue is further mitigated by using an asymmetric correlator receiver whose effectiveness depends on the excellent autocorrelation properties of the code.

We demonstrate the use of a modified P₄ polyphase code (P₄₃), originally developed for RADAR [10], for distributed acoustic sensing with an integrated coherent transceiver for telecommunication applications. Experimental results show that the proposed P₄₃ code significantly improves crosstalk suppression, achieving gains of 24 dB and 17.4 dB relative to single-sideband and double-sideband linear frequency-modulated probes (SSB-LFM and DSB-LFM), respectively. Moreover, co-directional propagation of the P₄₃ probe with a data signal at 100 GHz frequency spacing in a 50 km link leads to negligible impact on the transmission performance. Results highlight an increase of the required optical signal-to-noise ratio (rOSNR) by only 0.09 dB for 200 Gbit/s DP-16QAM and 0.06 dB for 100 Gbit/s DP-QPSK. This is lower than the maximum impact allowed for counter-directional integration defined in ITU-T G.681 [11].

Experimental setup

Fig. 1a shows the Coherent Correlation OTDR (CC-OTDR) setup with a telecommunication-optimised coherent optical subassembly (COSA). A continuous wave signal from a fibre laser with a Lorentzian linewidth of approximately 100 Hz is

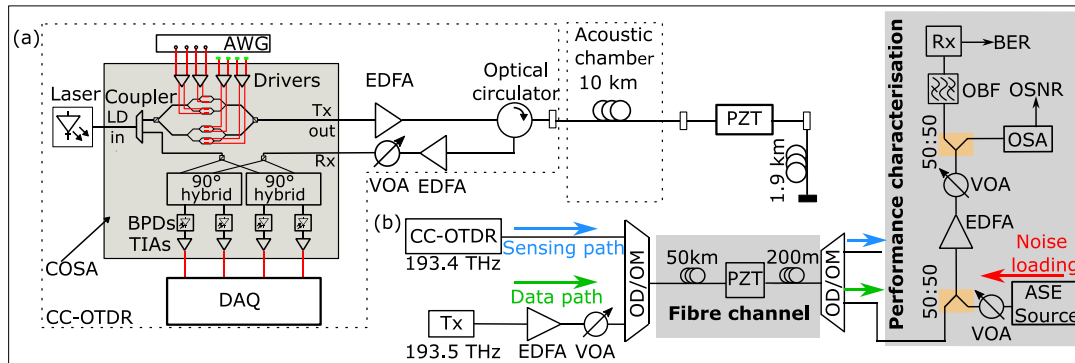


Fig. 1: a) CC-OTDR experimental setup, b) Schematic of the experimental setup for co-directional joint communication and sensing using CC-OTDR and coherent transceiver

fed into the laser input of the COSA. The power of the light is split via a coupler into two paths: one for the coherent receiver as local oscillator (LO) and the other for signal modulation, which is done by a dual-polarisation I/Q modulator. A two-channel arbitrary waveform generator modulates the digitally generated SSB-LFM, DSB-LFM, and P_4 probe on the optical signal. Both SSB-LFM and DSB-LFM have a probe signal duration of $8.192 \mu\text{s}$ and a chirp bandwidth of 100 MHz, while the P_4 code consists of 1024 bits at 125 MBaud, leading to the same duration. To prevent probe signal overlap, a zero-padding of $146 \mu\text{s}$ duration was added, resulting in a repetition frequency of 6.48 kHz. Subsequently, the signals are amplified by an erbium-doped fibre amplifier (EDFA) and fed into a 12 km fibre via an optical circulator at a peak power of 15 dBm. A vibration is generated by means of a piezoelectric transducer (PZT) with a frequency of 80 Hz and dynamic strain of $8.7 \text{ n}\epsilon$ over 15 m fibre. A second EDFA amplifies the backpropagating signal, which is then passed through a variable optical attenuator (VOA) and received at the Rx port of the COSA. Here, the four orthogonal electrical field components are extracted and subsequently sampled with 625 MS/s by a data acquisition (DAQ) card for further offline processing.

The joint communication and sensing (JCAS) setup with co-directional signal propagation can be observed from Fig. 1b. A dual-polarisation coherent data signal at 193.5 THz from an optical transmitter (TX), along with probe signal from a CC-OTDR at 193.4 THz, are multiplexed via an optical demultiplexer/multiplexer (OD/OM) and launched into the fibre with an average power of 0 dBm and a peak power of $\sim 10.5 \text{ dBm}$, respectively. The data signal was generated using DP-QPSK and DP-16QAM formats at a data rate of 100 Gbit/s and 200 Gbit/s, respectively. For sensing a 50 km fibre, the SSB-LFM with a longer duration of $16.384 \mu\text{s}$ and 100 MHz chirp bandwidth, and 2048 bits P_4 code at 125 MBaud was used. To prevent signal overlap, a silent window of 0.59 ms was incorporated between the probing sequences. The data signal is demultiplexed via an OD/OM, and noise loading is performed by adding amplified spontaneous emission (ASE). The optical spectrum analyser (OSA) is employed to measure the OSNR. The signal is then passed through a filter to eliminate out-of-band ASE noise and then detected at the receiver, which provides pre-forward error correction (FEC) bit error ratio (BER) measurement and identifies any uncorrected data blocks.

Results

To evaluate the different probing signals, the

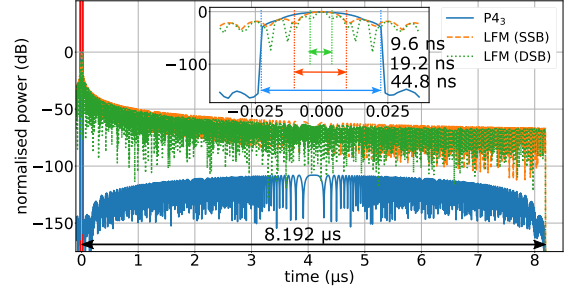


Fig. 2: Simulated pulse compression result of the three digital probing signals used in the experiment.

peak sidelobe (PSL) ratio metric, commonly used in RADAR applications, was employed as described in [4]. We first compare the simulated pulse compression properties of the probing sequences in ideal conditions, followed by experimental validation and crosstalk analysis. Fig. 2 shows the ideal pulse compression result of the three digital probe signals used in the CC-OTDR measurement for sensing. Herein, the normalised power as a function of time is plotted. The duration of the main peak for the SSB-LFM signal is 19.2 ns, and for the DSB-LFM signal, it is 9.6 ns, as illustrated in the inset. The main peak of the P_4 signal is broader than the others, with a duration of 44.8 ns, but it achieves the highest PSL, with a value of approximately 108 dB, indicating significantly stronger sidelobe suppression compared to SSB-LFM (13.7 dB) and DSB-LFM (8.1 dB).

In Fig. 3, we present the measured pulse compression result for a signal reflected from an open physical contact LC connector, with a dust cap, at the fibre end. The inset plot highlights the main peak, showing reflections from all three probing sequences. As observed from the figure, the P_4 code shows the highest PSL of 36.5 dB and the broadest peak duration of 44.8 ns. DSB-LFM has the lowest PSL of 10.1 dB and the narrowest peak of 9.6 ns. A PSL of 13.5 dB was measured for the SSB-LFM, and it exhibits a width of 20.8 ns. This highlights an inherent trade-off: while the broader main lobe of the P_4 code reduces spatial resolution, its strongly suppressed sidelobes significantly reduce inter-spatial crosstalk, thereby reducing event spread over the fibre length.

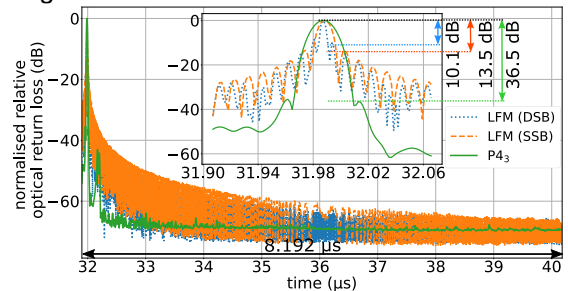


Fig. 3: Measured pulse compression result showing the fibre end reflection.

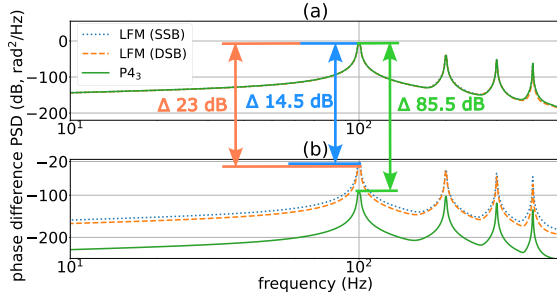


Fig. 4: Simulated phase difference PSD: a) location of the event, b) 10 m after the event.

Fig. 4 illustrates the phase difference power spectral density (PSD) at two locations along the fibre in a Rayleigh backscattering simulation model comparing the three probe sequences. The model simulates the reflections of scattering centres along the fibre in a similar approach to [12]. A 10 km fibre with a 100 Hz vibration event at 7 km was simulated. Laser phase noise, ASE noise, and receiver thermal noise were neglected to study the effect of sidelobe noise. Fig. 4 shows the mean PSD of 100 individual measurements. In Fig. 4a, at the event location, the PSD at 100 Hz is -5.9 dB (rad²/Hz) for all sequences, with visible harmonics. Fig. 4b shows the PSD 10 m after the event, where the SSB-LFM and DSB-LFM have a value of -20.4 dB (rad²/Hz) and -28.9 dB (rad²/Hz), respectively, at 100 Hz. The P4₃ code showed a lower PSD of -91.4 dB (rad²/Hz). From the results, it can be seen that the P4₃ code sees a gain of at least 62.5 dB in crosstalk suppression as compared to the LFM sequences.

Fig. 5 illustrates the phase difference PSD from experimental data at two locations, comparing the probing sequences. In Fig. 5a, the PSD at the PZT location shows a peak at 80 Hz, with SSB-LFM measuring -8.3 dB (rad²/Hz) and DSB-LFM at -8.6 dB (rad²/Hz), with a significant peak at the second harmonic. The P4₃ recorded a PSD of -6 dB (rad²/Hz). Fig. 5b shows the 80 Hz component is still present 10 m after the event location, with SSB-LFM at -31.9 dB (rad²/Hz) and -38.8 dB (rad²/Hz) for DSB-LFM. The P4₃ code has a value of -53.6 dB (rad²/Hz), limited by the noise floor level. From experimental results, the P4₃ code

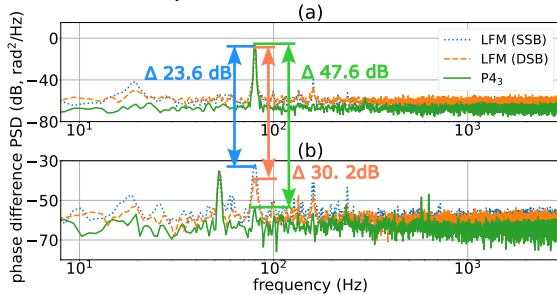


Fig. 5: Experimental phase difference PSD: a) location of the event, b) 10 m after the event.

sees a minimum gain of 17.4 dB in crosstalk suppression as compared to other tested sequences.

Tab. 1: Required OSNR for error free transmission.

Mode	rOSNR
DP-16QAM CC-OTDR off	19.13 dB
DP-16QAM CC-OTDR on with SSB-LFM	19.65 dB
DP-16QAM CC-OTDR on w/ shaped SSB-LFM	19.30 dB
DP-16QAM CC-OTDR on with P4 ₃	19.22 dB
DP-QPSK CC-OTDR off	11.87 dB
DP-QPSK CC-OTDR on with SSB-LFM	12.31 dB
DP-QPSK CC-OTDR on w/ shaped SSB-LFM	12 dB
DP-QPSK CC-OTDR on with P4 ₃	11.93 dB

The non-linear signal interaction impact of the three probe sequences in a JCAS scenario with a co-propagating data signal on the transmission performance has been evaluated using the setup depicted in Fig. 1b. Here, a 2048 bits P4₃ code, an SSB-LFM lasting 16.384 μ s, and an isosceles-trapezoidal-shaped SSB-LFM of the same duration with 4.096 μ s duration of the up- and down-ramp were compared. Tab. 1 shows the experimentally determined rOSNR to obtain a quasi-error-free transmission after FEC decoding. It shows that with CC-OTDR being switched on, the P4₃ code has a negligible impact on the transmission performance, resulting in an increase of only 0.09 dB for 200 Gbit/s DP-16QAM and 0.06 dB for 100 Gbit/s DP-QPSK. Due to its smooth power envelope, fast power transients leading to cross-phase modulation-induced phase jumps in the data signal are avoided. In contrast, with CC-OTDR switched on, probing with SSB-LFM leads to a more substantial penalty of 0.52 dB for 200 Gbit/s DP-16QAM and 0.44 dB for 100 Gbit/s DP-QPSK. With the shaped SSB-LFM, fast power transients are avoided, slightly reducing the penalty.

Conclusions

This work demonstrates that the P4₃ code significantly reduces inter-spatial crosstalk from sidelobe noise, reducing event spread over the fibre length. Furthermore, its negligible impact on the transmission performance of a co-propagating data channel is highlighted. Simulation results show that the P4₃ code provides higher crosstalk suppression, achieving improvements of 71 dB and 62.5 dB compared to SSB-LFM and DSB-LFM probing, respectively. Experimental measurements yielded values of 24 dB and 17.4 dB, bounded by the measurement noise floor. In the tested joint communication and sensing scenario with co-directional signal propagation over 50 km at 100 GHz signal spacing, the P4₃ code introduces a negligible transmission performance penalty, increasing the rOSNR by only 0.09 dB and 0.06 dB for 200 Gbit/s DP-16QAM and 100 Gbit/s DP-QPSK, respectively.

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